

Week 4 · Lesson 1 — React Fundamentals

Time: ~6–7h · **Goal:** Learn React's component model — the "small building blocks built on top of each other" your manager wants. Components, JSX, props, state, and the core hooks.

Why this matters for EVB-01

The current UI is one 45KB HTML file + one 110KB JS file — hard to maintain. React lets you split it into reusable pieces (`<StatusBar/>` , `<SensorConfigCard/>` , `<StartButton/>`) that manage their own state and re-render automatically when data changes. That's the whole reason for the rewrite.

Core concepts

- **Component:** a JavaScript function that returns UI (JSX). Compose small components into bigger ones.
- **JSX:** HTML-like syntax inside JS. `className` instead of `class` ; `{expression}` to embed JS values.

```
jsx function BatteryBadge({ percent }) { return <span className="badge">{percent}%</span>; }
```
- **Props:** inputs passed *into* a component (read-only), like function arguments. `<BatteryBadge percent={87} />` .
- **State (`useState`):** data a component owns that can change over time; changing it re-renders the component.

```
jsx const [recording, setRecording] = useState(false);
```
- **Effects (`useEffect`):** run side effects (fetch data, open a WebSocket) after render, and clean them up. You'll use this to connect to the device.
- **Lists & keys:** render arrays with `.map()` and a unique `key` prop.
- **Events:** `onClick={handleClick}` — same events as plain JS, just attached in JSX.
- **"Thinking in React":** break a design into a component tree, decide what's props vs state, lift shared state up to a common parent.

Watch / read (in order)

- **[CORE]** react.dev — *Quick Start / Learn React*: <https://react.dev/learn> — the official, modern (hooks-based) tutorial. Work through it.

- **[CORE]** react.dev — *Thinking in React*: <https://react.dev/learn/thinking-in-react> — directly teaches breaking a UI (like the dashboard) into components.
- **[CORE]** react.dev — *Tutorial: Tic-Tac-Toe*: <https://react.dev/learn/tutorial-tic-tac-toe> — hands-on state practice.
- **[OPTIONAL]** Scrimba — *Learn React* (free, interactive, by Bob Ziroll): [https://scrimba.com/learn-react-c0e](https://scrimba.com/learn/react-c0e) — great if you prefer guided video + code.

Practice (do this) → push to git

(You'll set up the build tool in Lesson 2; for now use the in-browser sandbox on react.dev or CodeSandbox.)

1. Build a `<SensorConfigCard>` component with `useState` for each setting (ODR, range, duration) and a `<StartButton>` whose label toggles between "Start" and "Stop".
2. Pass the battery % into a `<StatusBar>` via props.
3. Commit your sandbox code into `week-4/react-practice/` and push.

Before you move on, you should be able to...

- [] Write a component that takes props and returns JSX.
- [] Use `useState` to toggle something on click.
- [] Explain props vs state, and sketch the dashboard as a component tree.